

Ethanol Blending in Gasoline - Jamaica

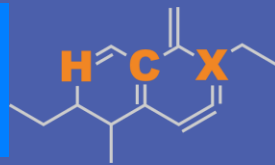
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**U.S. GRAINS &
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



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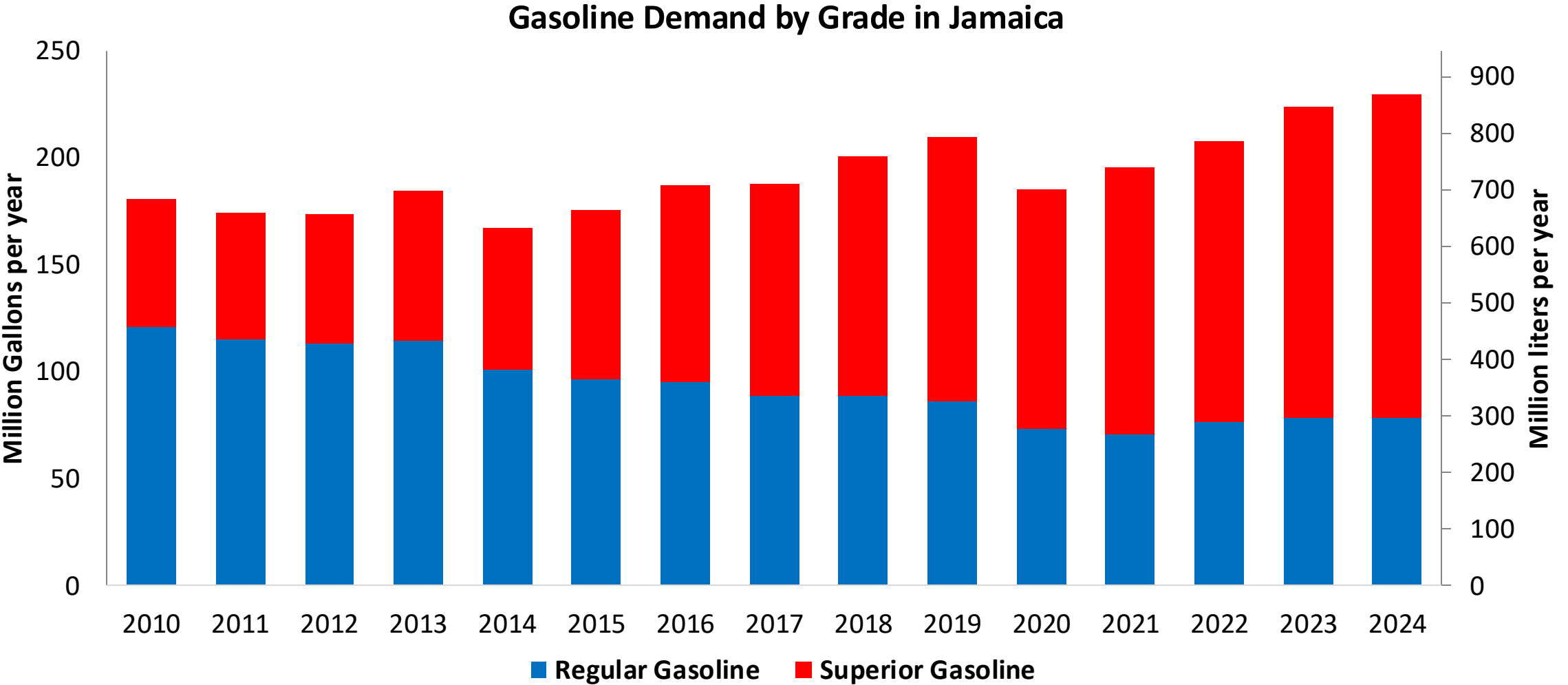


In 2024, gasoline demand was 230 million gallons (869 million liters). Gasoline production was 34 million gallons (130 million liters) and imports 195 million gallons (739 million liters) mainly from United States. Regular grade (AKI 87) had a market share of 34% and Premium grade (AKI 90) 66%.

Current gasoline quality norm mandates E10 (JS 341:2017) . Previously, ethanol was supplied with local production to Petrojam, however, in recent years, ethanol is imported from the United States.

Source: Petrojam, MSET

Gasoline Demand in Jamaica



Source: Petrojam, MSET

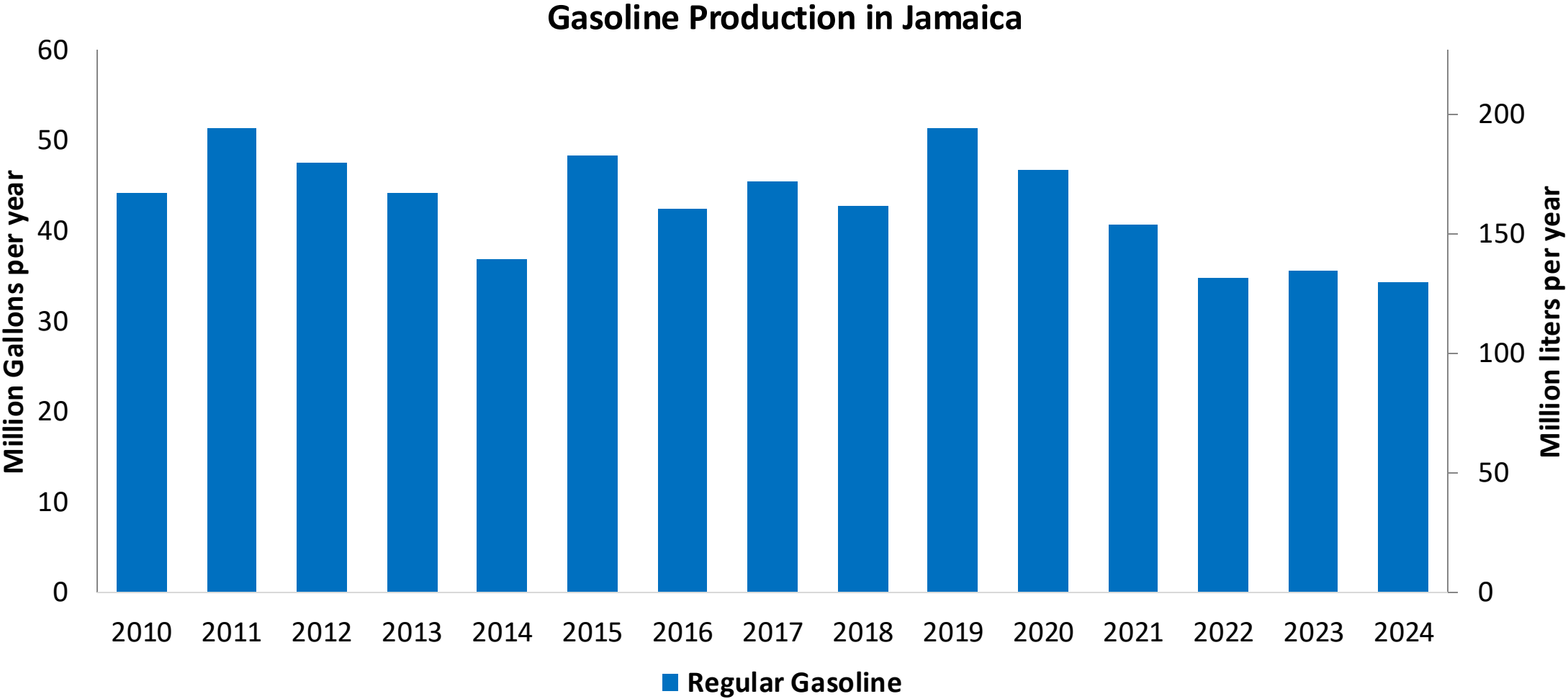
The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexadiene derivative. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.

This stacked bar chart displays the annual sales of Regular Gasoline and Superior Gasoline from 2010 to 2024. The left Y-axis measures sales in Million Gallons per year (0 to 250), and the right Y-axis measures sales in Million liters per year (0 to 900). Regular Gasoline is represented by blue bars, and Superior Gasoline is represented by red bars. The total sales for each year are the sum of the two components.

Year	Regular Gasoline (Million Gallons)	Superior Gasoline (Million Gallons)	Total Sales (Million Gallons)
2010	78	60	138
2011	65	58	123
2012	66	61	127
2013	71	69	140
2014	65	66	131
2015	48	80	128
2016	53	93	146
2017	44	99	143
2018	47	112	159
2019	36	123	159
2020	28	111	139
2021	31	124	155
2022	43	131	174
2023	43	146	189
2024	45	151	196

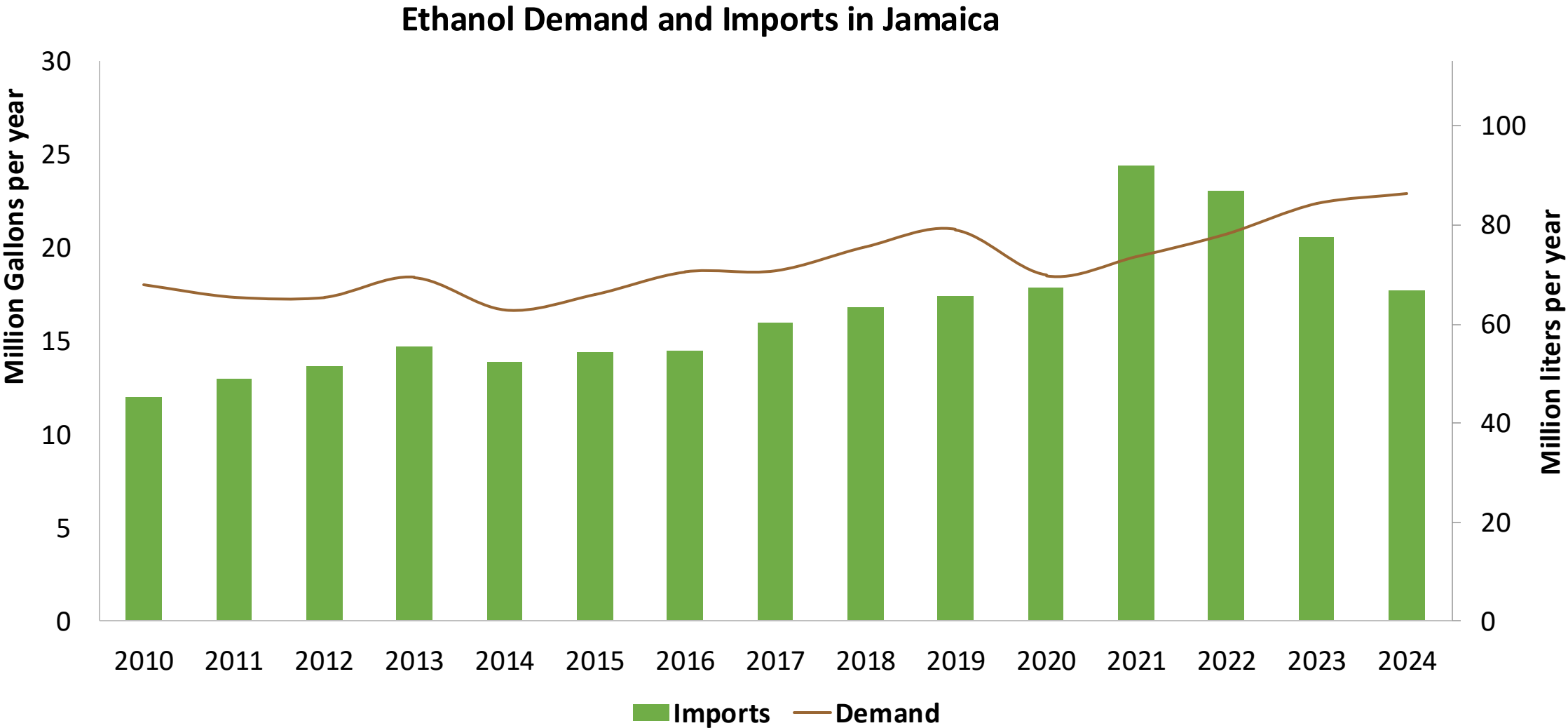
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Gasoline Production in Jamaica



Source: Petrojam, MSET

Ethanol Balance in Jamaica



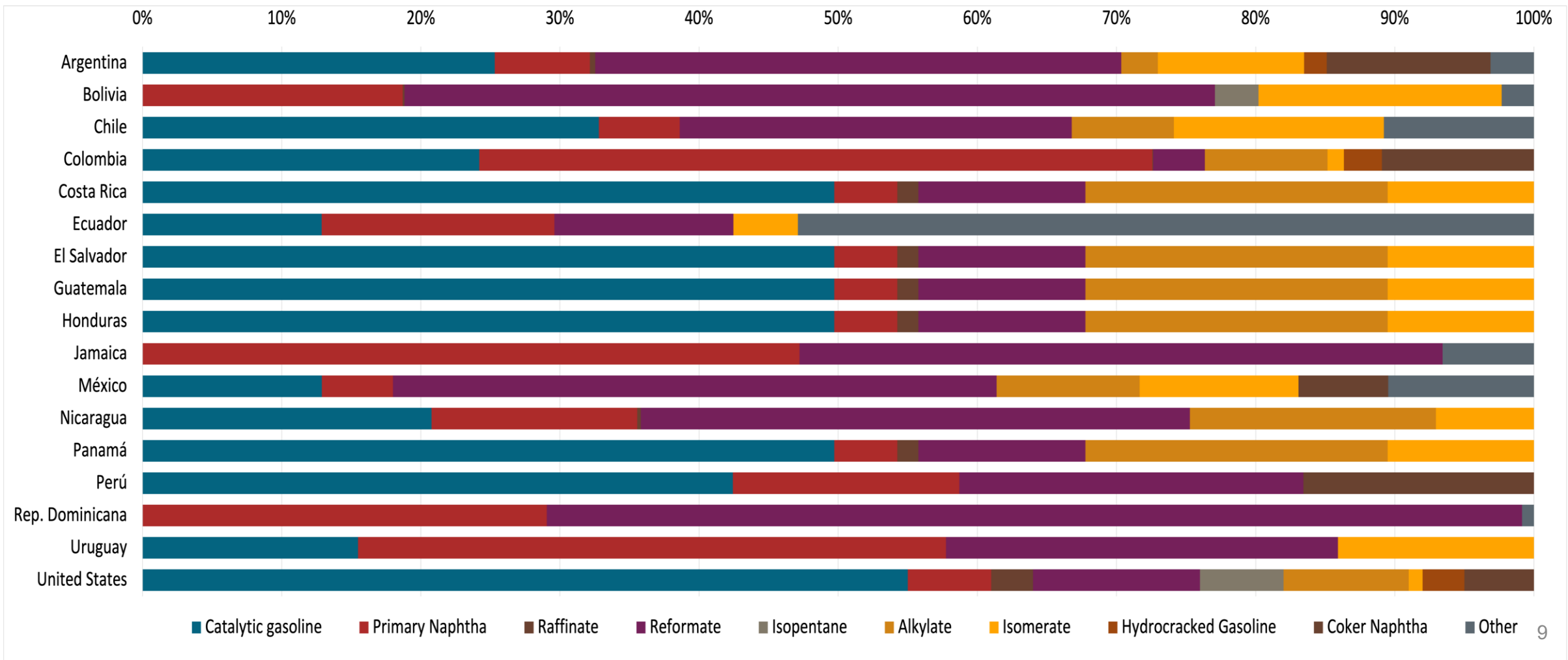
Source: MSET

Gasoline Quality in Jamaica

Name	JS 341:2017		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2017		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Unleaded Petrol AKI 87 E10	Unleaded Petrol AKI 90 E10	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 5 %v/v	< 5 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 45 %v/v	< 45 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 18 mg/l	< 18 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON			> 95	> 95	> 98	> 98
MON			> 85	> 88	> 85	> 88
AKI	90	87				
Sulfur Content	< 1.500 mg/kg	< 1.500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 4 %m/m (Gasoline with no ethanol)	< 4 %m/m (Gasoline with no ethanol)	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 61 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	< 15% v/v for Gasoline with no ethanol	< 15% v/v for Gasoline with no ethanol	-	-	-	-
Ethers 5 or more C Atoms	15% v/v (other oxygenates)	15% v/v (other oxygenates)	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



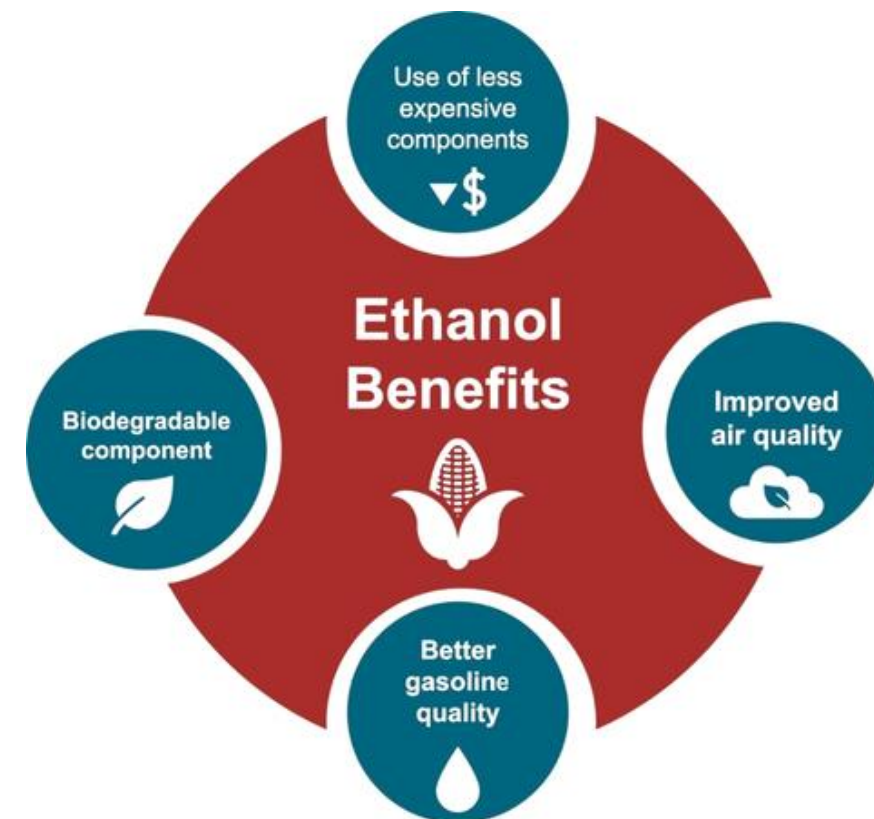
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

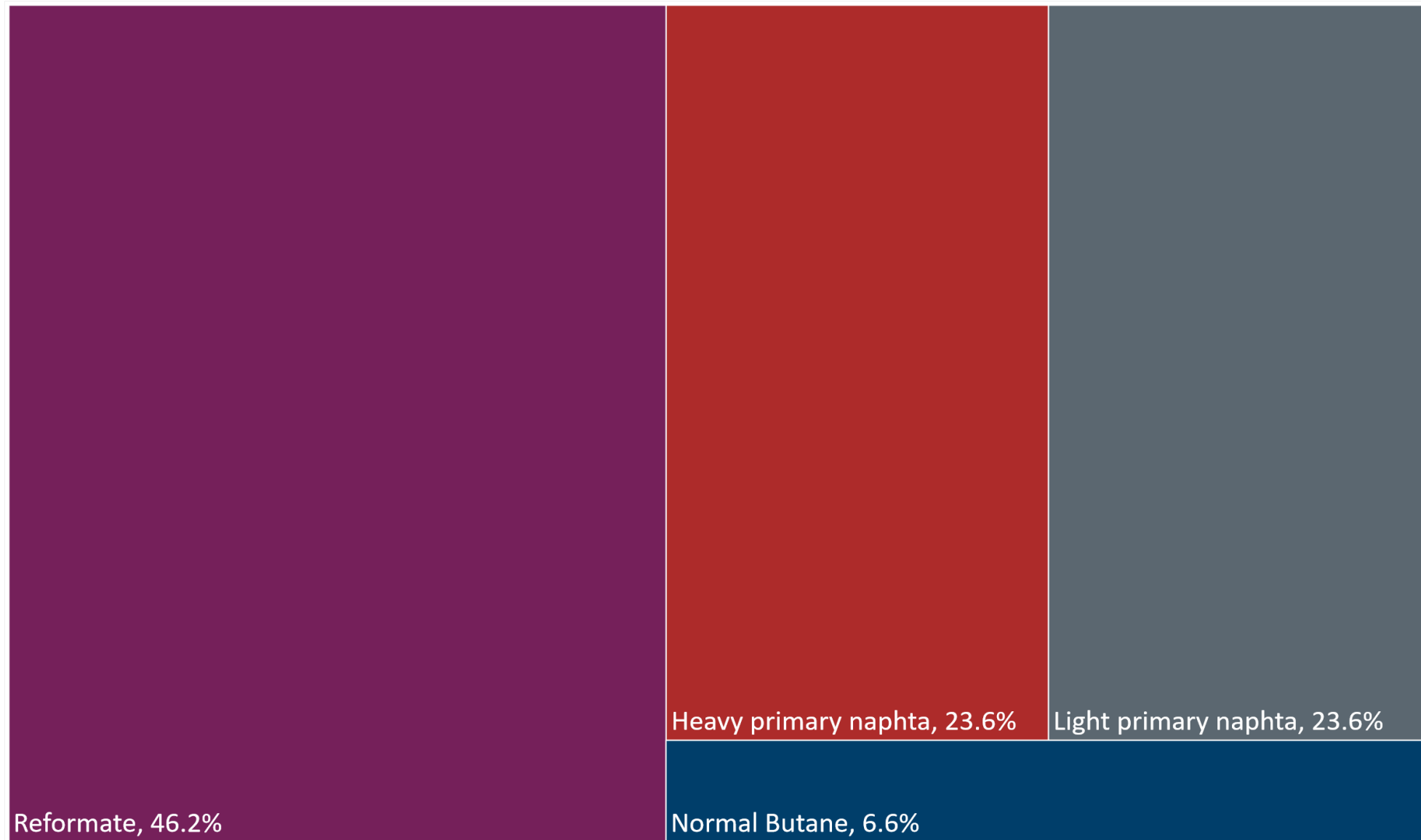
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

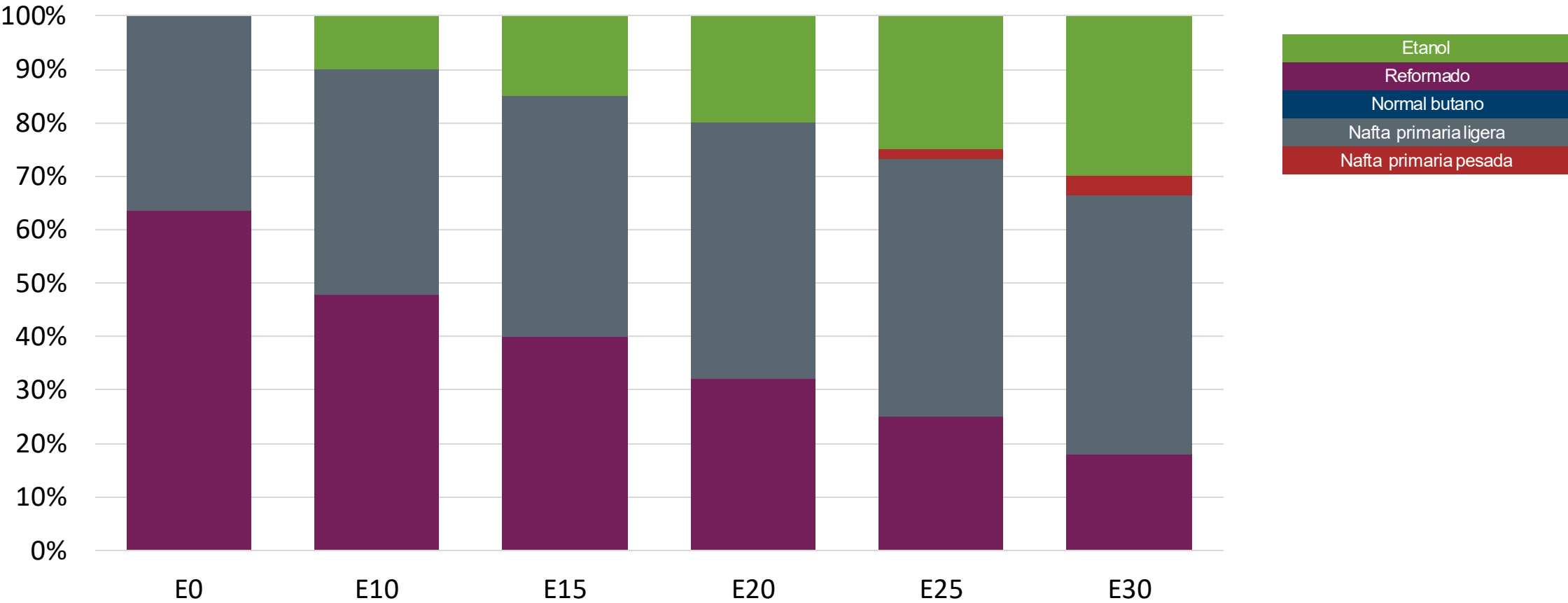
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components



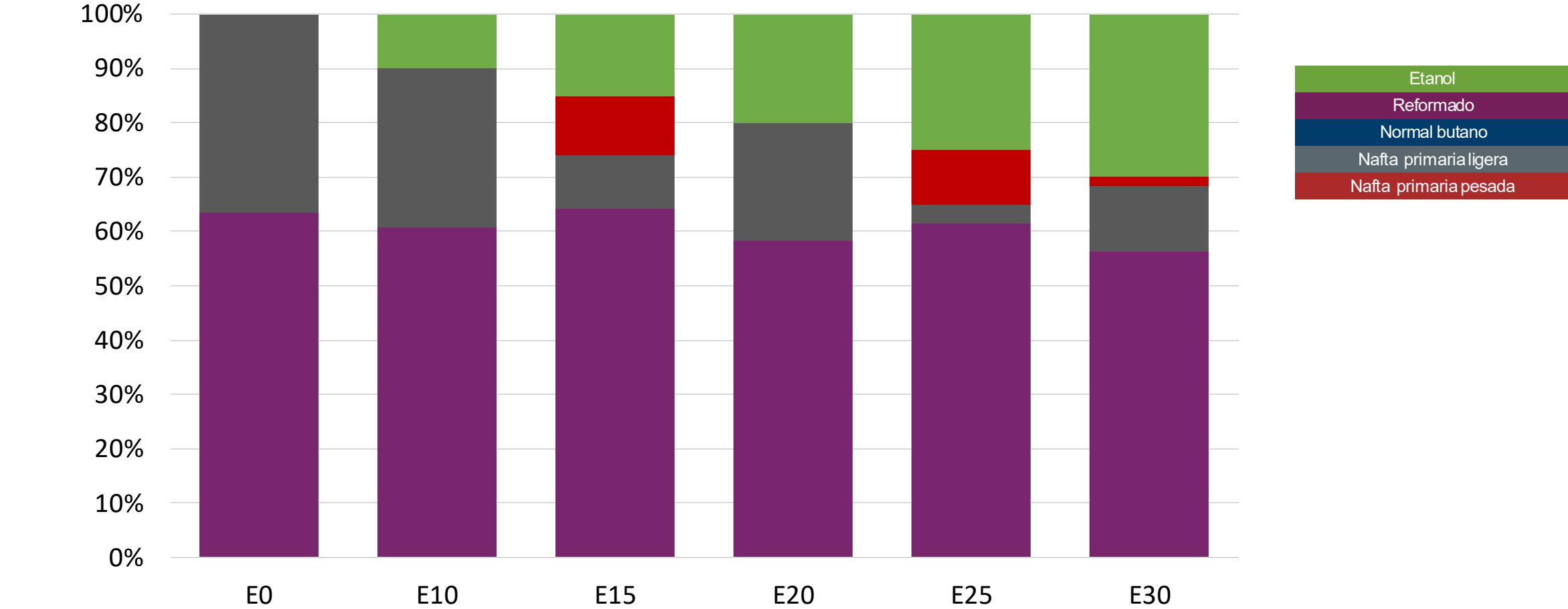
Regular Gasoline – Constant Octane



Octane (AKI)	87.0	87.0	87.0	87.0	87.0	87.0
Price (USD/gal)	\$ 2.512	\$ 2.314	\$ 2.214	\$ 2.115	\$ 2.016	\$ 1.917

Source: Faro90

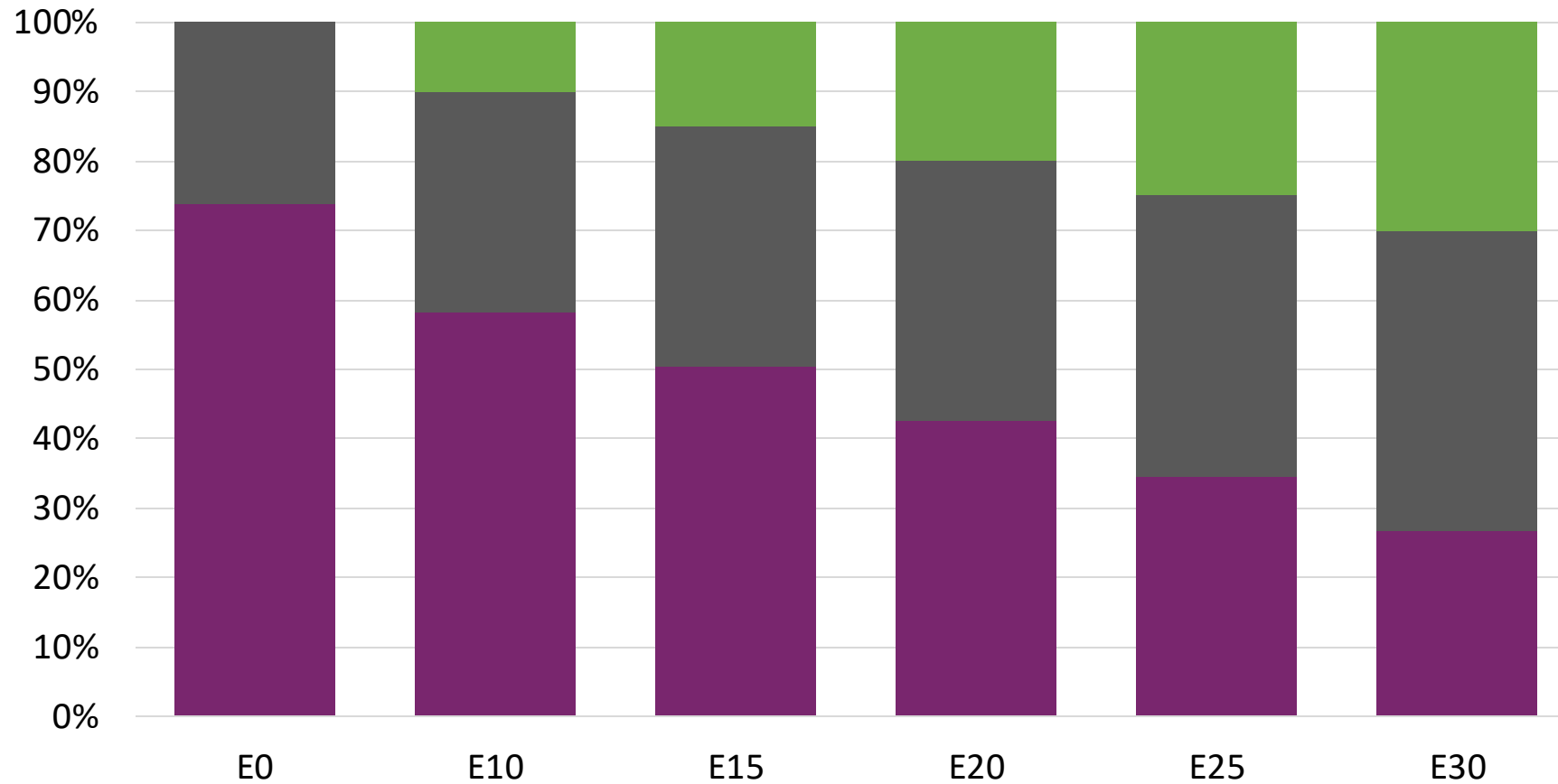
Regular Gasoline – Increased Octane



Octane (AKI)	87.0	90.8	92.6	94.5	96.4	98.3
Price (USD/gal)	\$ 2.512	\$ 2.512	\$ 2.512	\$ 2.512	\$ 2.512	\$ 2.512

Source: Faro90

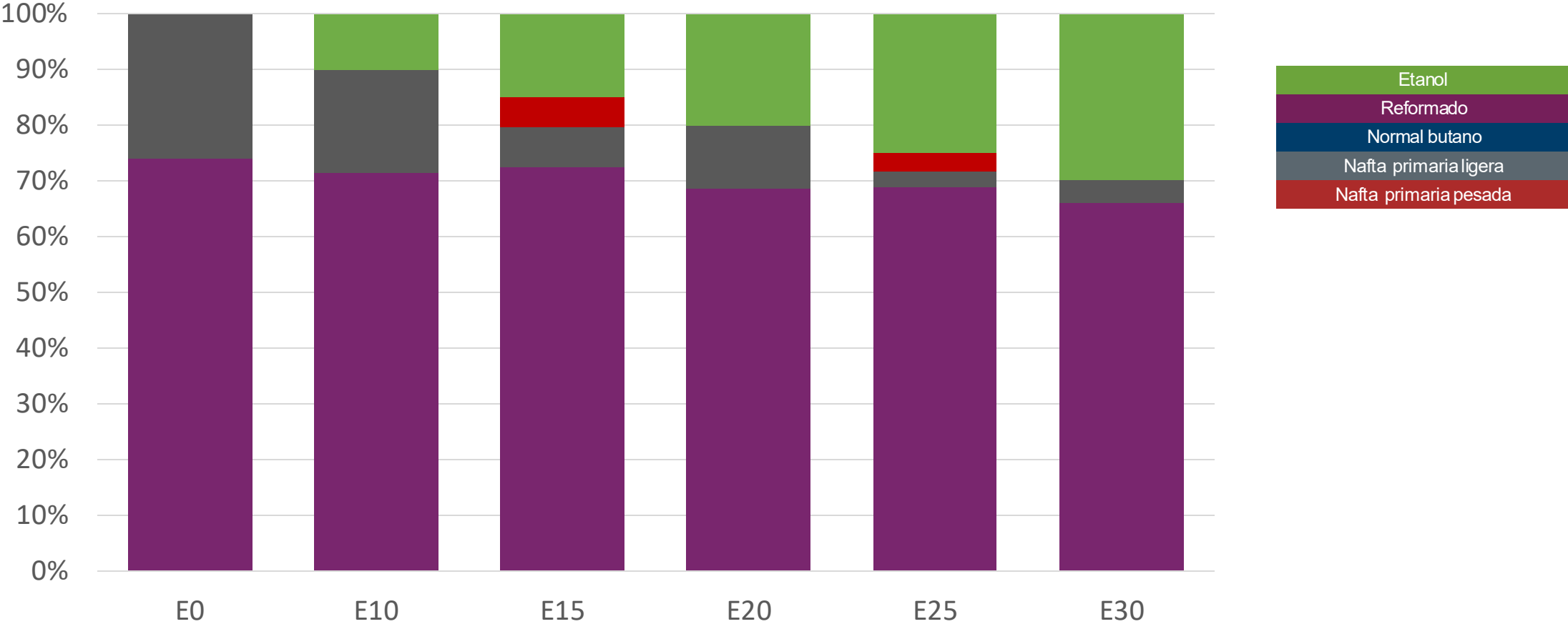
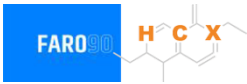
Premium Gasoline – Constant Octane



Octane (AKI)	90.0	90.0	90.0	90.0	90.0	90.0
Price (USD/gal)	\$ 2.670	\$ 2.472	\$ 2.372	\$ 2.273	\$ 2.174	\$ 2.075

Source: Faro90

Premium Gasoline – Octane Increment



Octane (AKI)	90.0	93.8	95.6	97.5	99.4	101.3
Price (USD/gal)	\$ 2.670	\$ 2.670	\$ 2.670	\$ 2.670	\$ 2.670	\$ 2.670

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

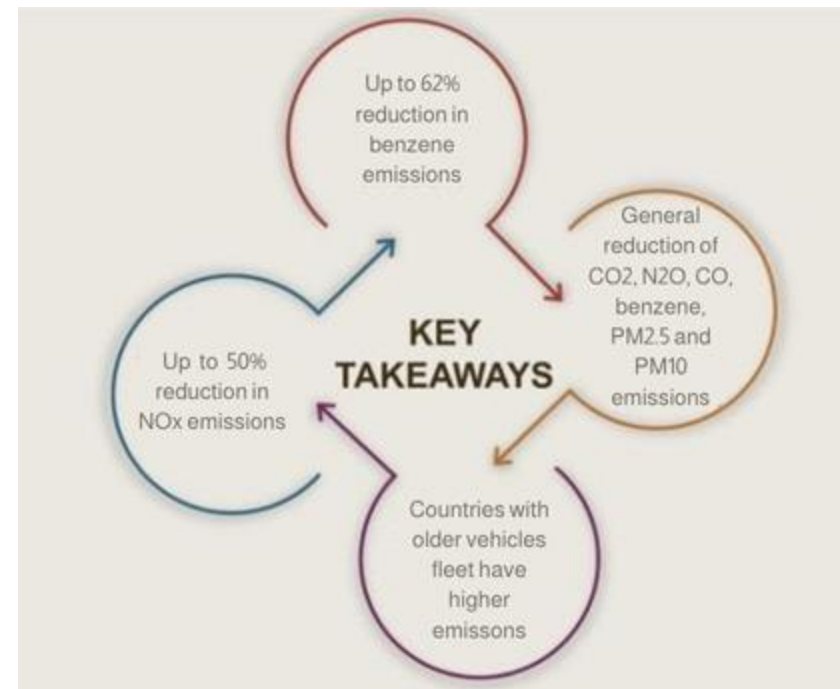
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

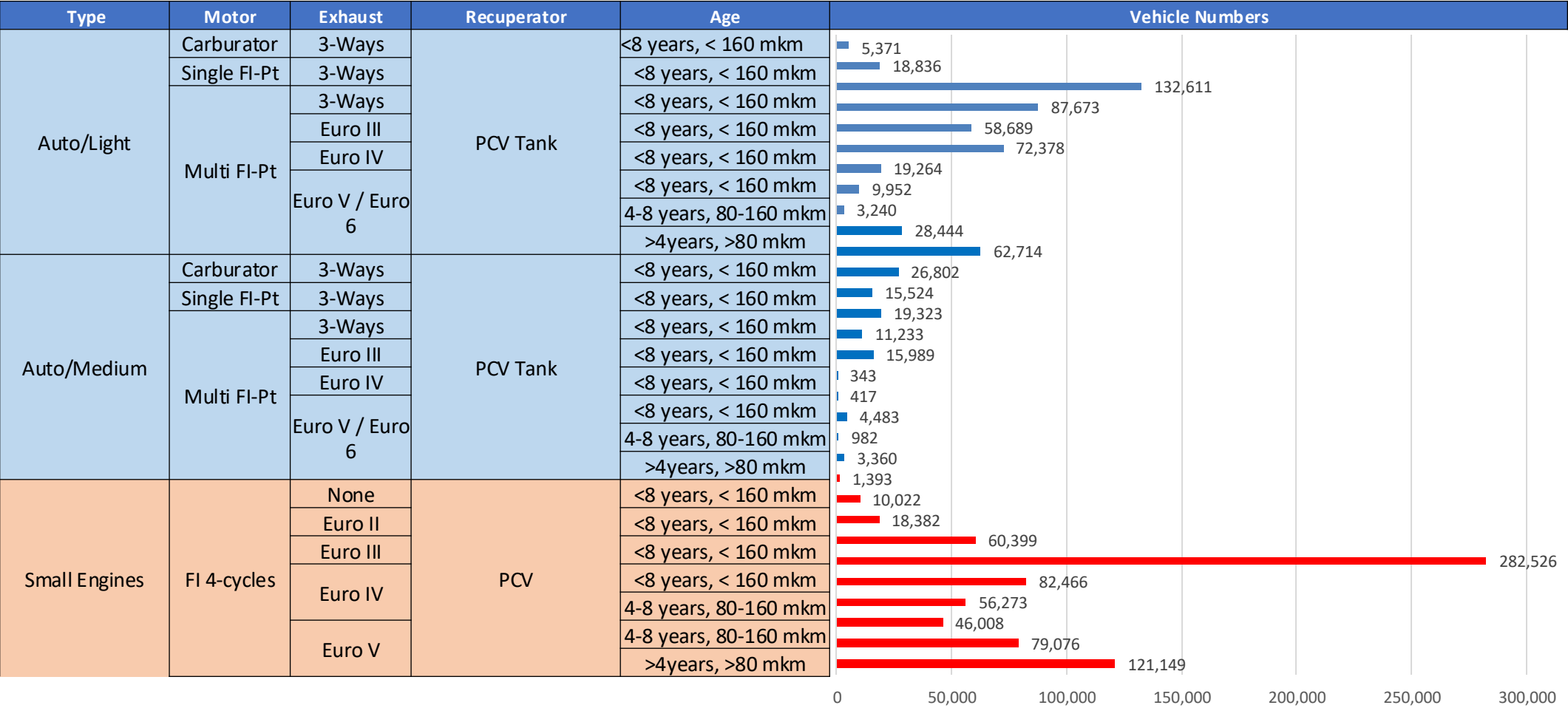
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



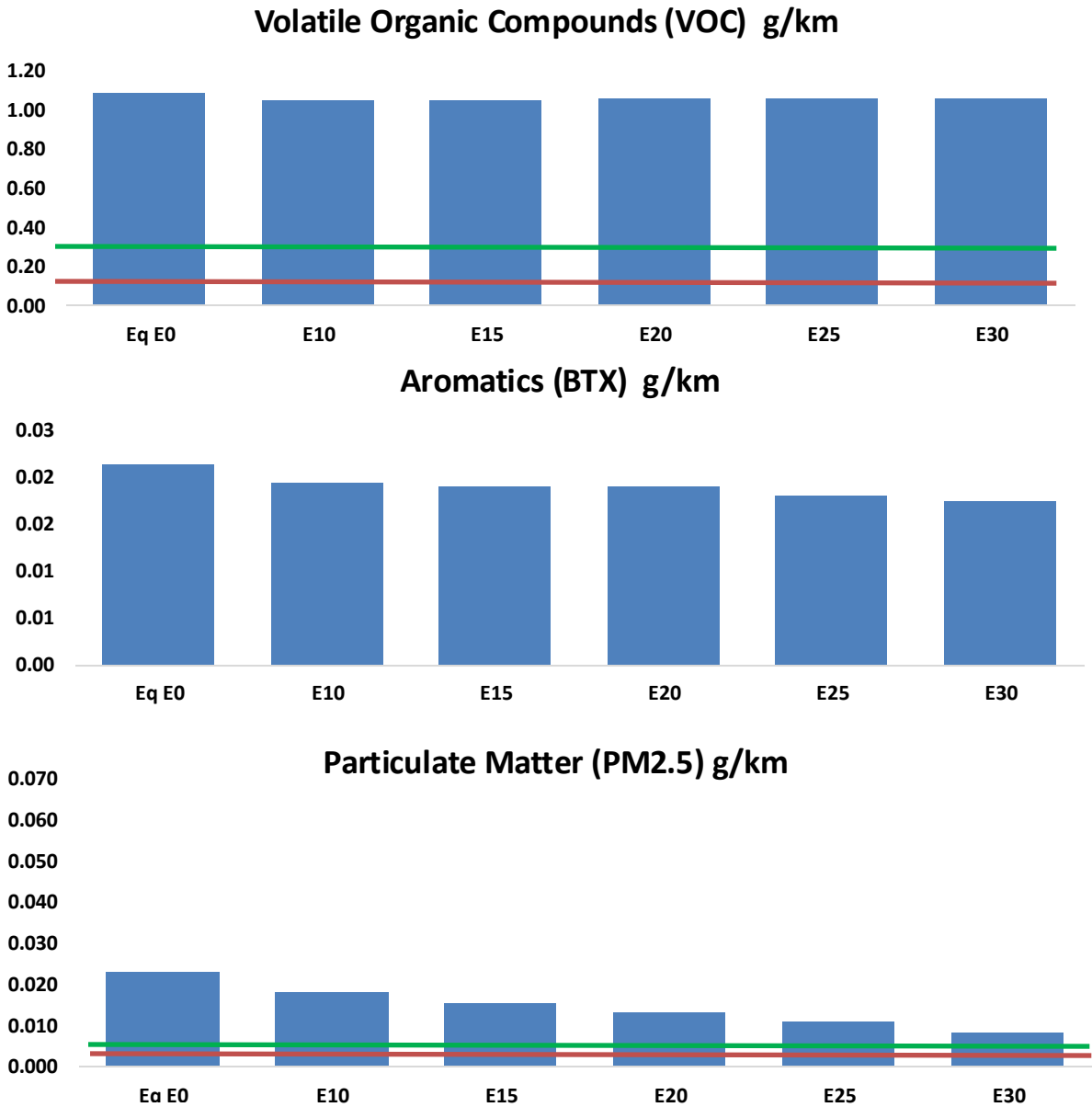
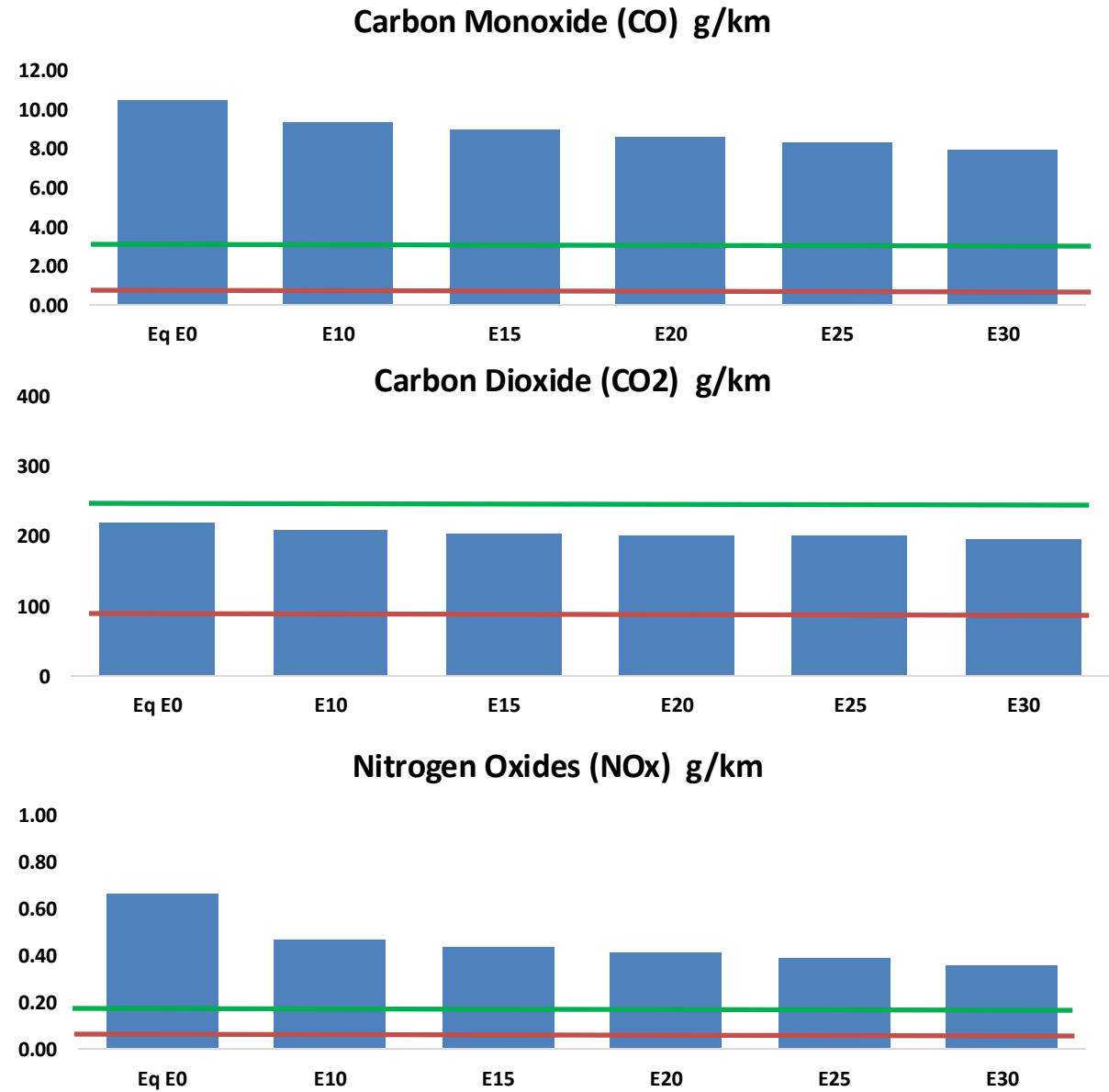
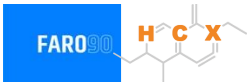
Gasoline Vehicle Fleet - Jamaica



Vehicular Fleet: **609,044**
Average Age: **20 years**
Motorcycles: **2.3%**

Jamaica – Vehicular Emissions

United States
Euro 6



Jamaica – Gasoline Vehicle Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	166,897	158,242	149,588	143,311	137,369	132,914	127,041	-10%	-18%
CO2	3,490,486	3,400,730	3,315,961	3,249,293	3,216,303	3,199,172	3,140,202	-5%	-8%
VOC	17,390	17,102	16,815	16,784	16,847	16,924	16,926	-3%	-3%
NOx	10,588	7,941	7,412	6,986	6,588	6,150	5,686	-30%	-38%
SOx	43	39	36	34	31	28	25	-15%	-28%
NH3	1,182	1,170	1,159	1,164	1,177	1,187	1,194	-2%	0%
N2O	55	54	54	54	55	56	57	-1%	2%
CH4	3,735	3,735	3,735	3,791	3,873	3,944	4,011	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	95	92	89	88	86	86	84	-6%	-9%
Acetaldehyde	175	201	294	448	611	698	824	68%	249%
Formaldehyde	501	487	568	667	699	773	841	13%	39%
Benzene	339	326	308	304	301	288	279	-9%	-11%
PM2.5	210	187	163	142	120	98	74	-22%	-43%
PM10	155	138	120	104	89	72	55	-22%	-43%

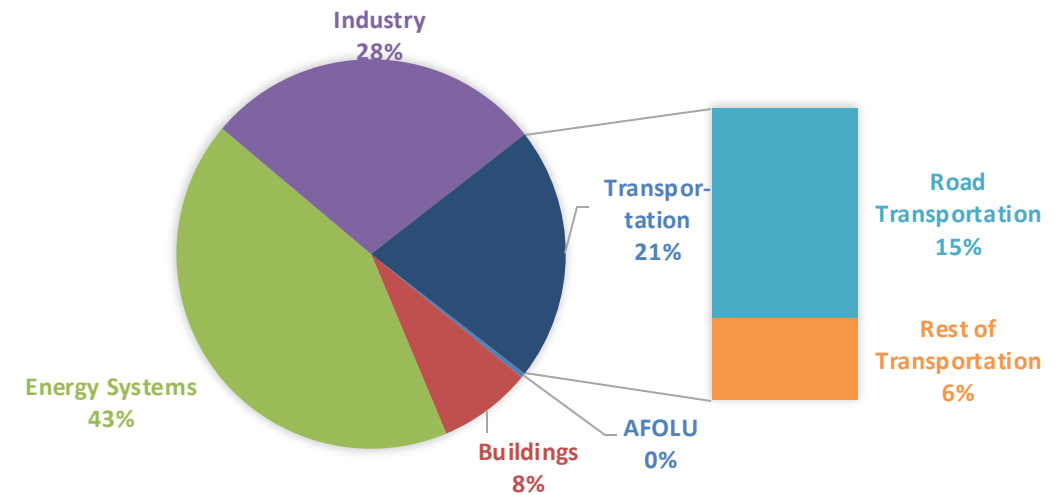
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

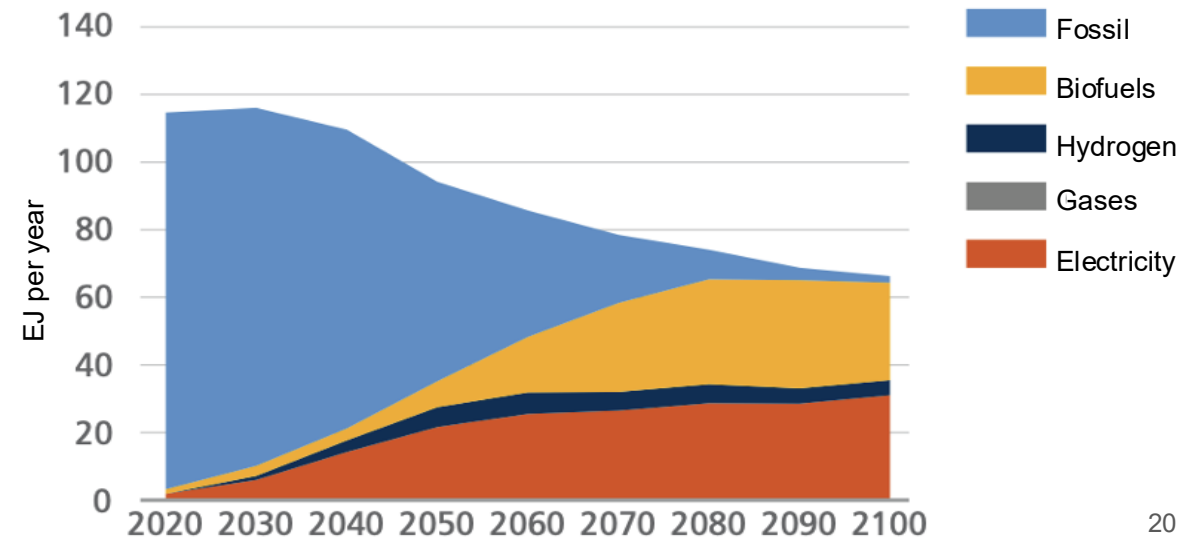
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

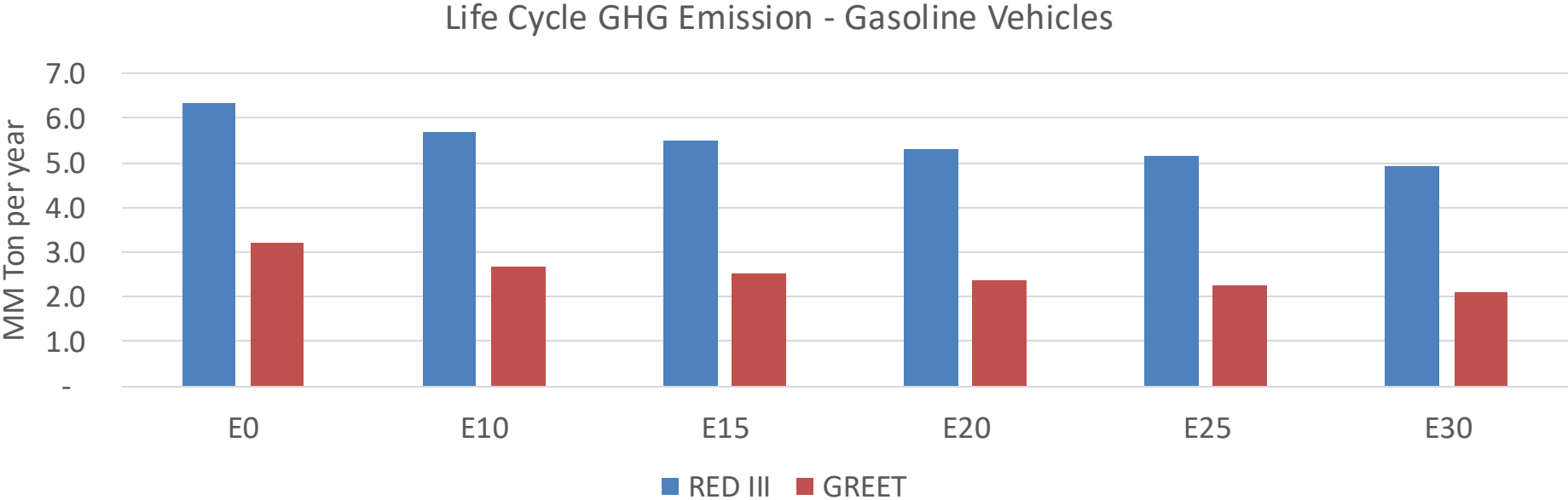
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Jamaica – Gasoline Vehicle Fleet GEI Emissions



Emisiones País 2024 (MMT/año)	9.4
Meta a 2035 (MMT/año)	4.7
Delta	4.7

%Reducción vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	17.2%	21.7%	25.9%	30.1%	34.9%
RED II/III	0.0%	10.3%	13.4%	16.2%	19.0%	22.0%

%	E0	E10	E15	E20	E25	E30
Meta@2035						
GREET 2024	0.0%	11.8%	14.9%	17.7%	20.6%	23.9%
RED II/III	0.0%	13.9%	18.0%	21.8%	25.5%	29.6%